Practical-1

1. Aim:

Create different blinking patterns with LEDs by connecting to GPIO pins of Raspberry Pi.

2. Prior Concepts:

i) Python Programming - loops, time delay, GPIO library

ii) Digital Electronics - LED works on HIGH (1) = ON, LOW (0) = OFF

3. New Concepts:

i) Raspi 4: It has 40 GPIO pins. You can control LEDs, sensors, motors from Python code.

ii) Hardware Interface: LED anode (+ve) goes to GPIO via 330 ohm resistor, cathode (-ve) to GND. Resistor protects LED from burning.

4. Procedure:

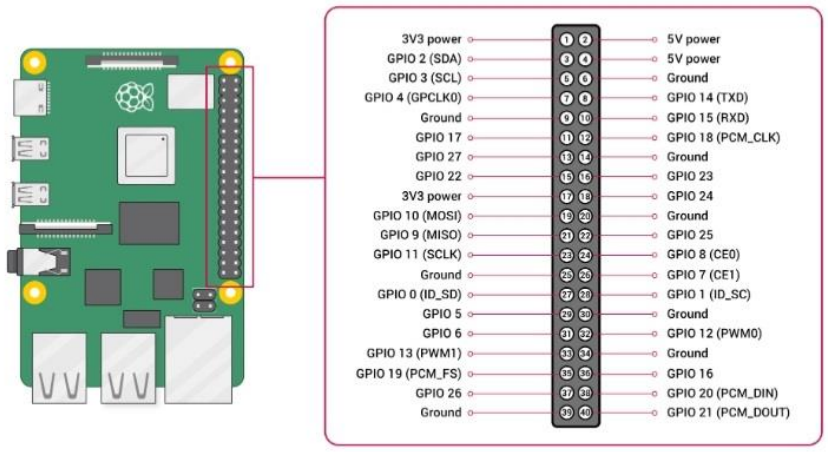
i) Install Python3 IDE:

Open terminal and type:

sudo apt install idle3

ii) Connection as per your diagram:

- Connect LEDs on breadboard
- Connect 330 ohm resistor to anode (long leg +ve) of LED
- Other end of resistor to GPIO pin (e.g., GPIO 17, 27, 22)
- Cathode (short leg -ve) to GND pin of Pi
- In your diagram: Right side is 40-pin header - connect as shown



5. Implementation with Python Code:

Question 1 : Write a program to generate TWO Bit binary Pattern.

Code :

```
import RPi.GPIO as GPIO
from time import sleep

GPIO.setmode(GPIO.BOARD)

led1=29  #Yellow
led2=31  #Green

GPIO.setup(led1,GPIO.OUT)
GPIO.setup(led2,GPIO.OUT)

GPIO.output(led1,False)
GPIO.output(led2,False)

def ledpattern(v1,v2):
    GPIO.output(led1,v1)
    GPIO.output(led2,v2)

try:
    while True:
        ledpattern(1,0)
        sleep(0.5)
        ledpattern(0,1)
        sleep(0.5)
        ledpattern(1,1)
        sleep(0.5)
        ledpattern(0,0)
        sleep(0.5)
```

finally:

GPIO.output(led1, False)

GPIO.output(led2, False)

GPIO.cleanup()

6. Results: